

Review

Review on Ansatz Architectures of Variational Quantum Algorithms for Continuous Optimization: From Fixed Structures to Adaptive Evolution

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Abstract

Variational quantum algorithms (VQAs) are a leading framework for realizing quantum advantages in the Noisy Intermediate-Scale Quantum (NISQ) era, with applications spanning discrete combinatorial problems and continuous optimization. While the topologies of parameterized quantum circuits (ansatzes) fundamentally govern both expressibility and trainability in continuous landscapes, existing reviews predominantly focus on static algorithmic classifications or discrete settings, leaving the structural evolution and practical limitations of ansatz architectures insufficiently explored. To address this gap, this review presents a systematic analysis of variational ansatz architectures, tracing their progression from static, pre-defined topologies to adaptive growth mechanisms. Beyond traditional gradient-driven and architecture-search paradigms, we evaluate supplementary strategies such as layerwise training and noise-adaptive construction, revealing inherent vulnerabilities such as local minima entrapment and the compilation overhead induced by calibration drift. The mathematical foundations of VQAs are outlined, and representative fixed ansatz architectures, including hardware-efficient, physics-inspired, and problem-specific designs, are characterized within continuous-domain mappings. Intrinsic limitations arising from barren plateaus (BPs) and noise-induced barren plateaus (NIBPs) are analyzed, revealing the fundamental coupling between circuit depth, parameter scaling, and trainability degradation. Furthermore, adaptive construction strategies and recent advances in automated variational quantum architecture search (VQAS) are examined. Through the synthesis of intrinsic limitations (BPs, NIBPs, and hardware-algorithm coupling) and the evaluation of standardized benchmarking protocols, this review rigorously assesses the resource trade-offs of current VQA frameworks. Ultimately, next-generation ansatz design will adopt hardware–software co-design principles grounded in physical constraints, enabling scalable and noise-resilient quantum optimization.

Keywords: variational quantum algorithms; continuous optimization; parameterized quantum circuits; adaptive ansatzes; barren plateaus; quantum architecture search



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1. Introduction

Quantum computing explores potential pathways toward advantage in addressing computational tasks that currently challenge classical computers [1,2]. However, substantial technical hurdles hinder the translation of this theoretical advantage into practical applications: stringent requirements for qubit scalability and error-rate thresholds constrain the realization of universal fault-tolerant quantum computers [2]. The field is currently

in the Noisy Intermediate-Scale Quantum (NISQ) era, where physical hardware typically comprises tens to hundreds of uncorrected noisy qubits. In this regime, gate infidelities and finite coherence times severely limit the executable depth of quantum circuits [2–4].

To mitigate these physical limitations, variational quantum algorithms (VQAs) have emerged as the primary candidates for achieving quantum advantage on near-term hardware [1,4]. VQAs rely on a hybrid quantum–classical framework that formulates problems as parameterized cost functions. In this paradigm, quantum computers prepare and measure quantum states, while classical optimizers iteratively update circuit parameters [1,5]. This framework adapts well to hardware constraints, reduces reliance on circuit depth, and shows inherent robustness against certain noise sources [6,7].

Within this hybrid optimization paradigm, VQAs have expanded to complex continuous optimization tasks [7]. In this review, we explicitly and exclusively define continuous optimization as problems defined on continuous domains (such as computational fluid dynamics [7], chemical process control, and continuous-variable energy landscapes) involving continuous decision variables. While discrete combinatorial problems like Max-Cut [8,9] serve as foundational baseline comparisons to illustrate the historical transition in topological design requirements, our scope strictly focuses on continuous mappings and parameterized cost landscapes. Unlike discrete problems, where decision variables map directly to Pauli operators, continuous problems require encoding real or complex variables into the amplitudes or phases of quantum states. This shift changes the underlying computational complexity. It is therefore important to distinguish whether an algorithm provides provable exact solutions or heuristic approximations when assessing its theoretical advantage [10]. For example, in computational fluid dynamics (CFD), mapping classical partial differential equations (PDEs) onto a variational framework can, in principle, achieve logarithmic scaling with respect to problem size in noise-free state-vector simulations [7]; however, this remains a severe empirical challenge on physical hardware due to depth and realistic noise constraints.

However, continuous optimization tasks typically involve non-convex cost landscapes, where algorithms can become trapped in local minima. When using fixed-structure ansatzes, capturing the dynamic features of continuous variables often requires increasing circuit depth and complexity [4,11]. As circuits grow deeper, they experience barren plateaus (BPs) due to excessive expressibility, as well as noise-induced barren plateaus (NIBPs) caused by hardware errors [12,13]. This exponential decay in cost-function gradients renders classical optimizers ineffective. As a result, the trade-off between expressibility and trainability under hardware constraints in static ansatzes remains a central bottleneck in quantum algorithm design.

The design of the variational ansatz plays a decisive role in the success of VQAs, as its circuit topology fundamentally governs both the expressibility of quantum state representations and the resulting trainability of the optimization landscape. While current review papers provide excellent broad overviews of quantum hardware [2], foundational VQA principles [1,4], or general quantum machine learning applications [14,15], they predominantly focus on static algorithmic classifications or remain heavily confined to discrete combinatorial tasks. This review does not seek to replace these foundational existing reviews, but rather to critically complement them. To achieve a balanced perspective, this review fills a unique and highly specific gap: it systematically traces the structural evolution of continuous-variable ansatzes from fixed topologies to dynamically adaptive architectures and rigorously maps the trade-offs between expressibility and trainability under realistic hardware constraints, providing an objective evaluation of their practical viability. Table 1 summarizes the differences between our review and prior literature across multiple dimensions, including review scope, optimization domain, and structural-evolution perspective.

Table 1. Comparison of This Review with Existing VQA Literature.

Reference	Primary Review Scope & Focus	Target Optimization Domain	Explores Ansatz Structural Evolution?	Addresses Hardware Noise & Trainability Trade-Offs?
[1]	General VQA framework and broad applications	Primarily Discrete/Quantum Chemistry	No (Focuses on static fundamental classifications)	Partial (General barren plateau discussion)
[2]	NISQ algorithms and hardware implementations	Hardware-agnostic/General	No (Focuses on hardware and algorithm typologies)	Yes (Hardware constraints focused)
[4]	VQA fundamental concepts and basic components	Mixed (Discrete heavily featured)	No (Classifies predefined algorithmic components)	Partial (General challenges)
[14,15]	Quantum Machine Learning (QML) applications	Data-driven/Machine Learning	No (Typically assumes fixed expressive ansatzes)	Partial (Focuses on QML expressibility limits)
This Work	Ansatz Architecture Design and Topology Evolution	Strictly Continuous Optimization	Yes (Traces progression from fixed HEA/QAOA to dynamic ADAPT/VQAS)	Yes (Explicitly maps expressibility vs. NIBP limits)

To address this gap, this paper presents a systematic review focused on variational ansatz architectures for continuous optimization. The core contributions are threefold. First, the topological features and inherent limitations of static ansatz architectures are systematically characterized, establishing a direct link between circuit depth, measure concentration, and susceptibility to barren plateaus (BPs) and noise-induced barren plateaus (NIBPs). Second, the structural evolution from static topologies to dynamically adaptive architectures is explicitly traced, with gradient-driven, geometry-aware, and automated search (VQAS) strategies evaluated for mitigating hardware constraints. Third, the trade-offs between expressibility and trainability across architectures are rigorously analyzed, bridging theoretical landscape insights with practical hardware benchmarking to establish a unified hardware–software co-design perspective for next-generation variational quantum circuits.

To ensure a rigorous and reproducible systematic review, a structured literature-search protocol was implemented. We queried databases including IEEE Xplore, Web of Science, and arXiv for literature published from database inception up to May 2026, using targeted keyword combinations such as “variational quantum algorithms”, “continuous optimization”, and “ansatz architectures”. The screening procedure consisted of two sequential stages: an initial title and abstract evaluation to filter out studies completely unrelated to quantum optimization, followed by a rigorous full-text assessment. Our inclusion criteria explicitly focused on studies addressing continuous-variable encoding and continuous optimization domains. Studies strictly confined to discrete combinatorial optimization without continuous-state mappings were excluded. Following this protocol, 81 core papers were selected. The proposed taxonomy was subsequently constructed by categorizing these papers based on their structural topology, systematically tracing the evolution from static, pre-defined structures to dynamic adaptive mechanisms.

Section 2 introduces the fundamental mathematical framework and standard execution of VQAs. Section 3 describes quantum representation methods, encoding strategies, and cost-function formulations for continuous optimization. Section 4 analyzes representative fixed-structure ansatzes and their inherent topological limitations in terms of expressibility and trainability. Section 5 presents dynamic adaptive growth mechanisms

guided by gradient and geometric information, reviews recent progress in automated variational quantum architecture search (VQAS), and compares these approaches in terms of resource trade-offs. Section 6 evaluates the limits of expressibility and trainability across architectures from a measure concentration perspective. Section 7 discusses physical implementation constraints and benchmarking practices on noisy quantum hardware. Finally, Section 8 summarizes the work and discusses the transition toward fault-tolerant quantum computing systems.

2. Fundamental Mathematical Architecture of VQAs

This section presents the general mathematical and execution framework of VQAs. This framework serves as the theoretical and operational foundation for the subsequent discussion of continuous-variable encoding and ansatz architecture design.

The general mathematical architecture of VQAs is built upon a hybrid quantum–classical paradigm, with its theoretical basis rooted in the Rayleigh–Ritz variational principle [5,16]. Unlike fully coherent quantum algorithms that require long circuit depths and fault-tolerant execution, VQAs distribute computational tasks between classical and quantum resources. This results in an iterative feedback loop that efficiently utilizes limited coherence time while partially mitigating the impact of hardware noise [1,2].

From a functional perspective, a VQA can be viewed as an optimization process over a parameterized family of quantum states $\{|\psi(\theta)\rangle\}$, where the objective is to minimize a cost functional defined on a high-dimensional Hilbert space. The optimization landscape is therefore implicitly determined by the joint structure of the ansatz, the encoding strategy, and the measurement operators. This coupling plays a central role in both expressibility and trainability, and will be revisited in subsequent sections.

The standardized execution of VQAs operates through a tightly coupled quantum–classical feedback loop, as illustrated in Figure 1. Following the preparation of an initial reference state $|\Phi_0\rangle$, a parameterized quantum circuit (or ansatz) $U(\theta)$ is applied to generate the trial state $|\psi(\theta)\rangle$. The cost function $C(\theta)$ is subsequently evaluated via quantum hardware measurements of target observables. Unlike binary Pauli mappings in discrete cases, navigating a continuous landscape introduces altered non-convex structures and heightened parameter sensitivity during classical optimization [10]. Furthermore, continuous domains demand distinct measurement strategies, as the statistical variance of continuous estimators significantly amplifies sample complexity [17,18]. A classical optimizer then iteratively updates the parameter vector θ based on these measurement outcomes [1,2,5]. To execute these gradient-based updates on physical hardware without relying on noise-sensitive finite differences, exact analytical gradients of the parameterized quantum circuits are typically evaluated using the parameter-shift rule [19].

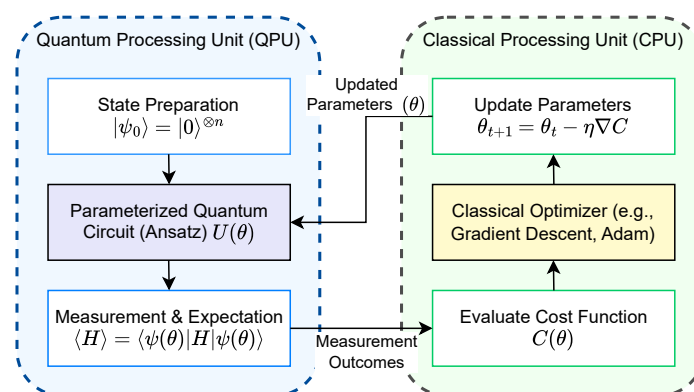


Figure 1. Schematic representation of the quantum-classical hybrid execution framework for variational quantum algorithms (VQAs).

Rather than merely serving as an algorithmic subroutine, the parameterization strategy and topology of $\mathcal{U}(\theta)$ constitute the central structural bottlenecks of the entire VQA pipeline. The choice of ansatz dictates the geometry of the accessible Hilbert subspace, which fundamentally governs the strict trade-off between expressibility and trainability [1,20]. Deep entangling circuits increase state-space coverage but rapidly amplify measurement costs and noise sensitivity [21], making topology design the primary determinant of hardware feasibility long before any classical optimization update occurs.

3. Quantum Representation Mechanisms for Continuous Optimization

Extending VQA frameworks to the continuous optimization domain requires specialized mathematical mappings between classical variables and quantum states. This section examines the computational complexity of continuous-variable problems, evaluates quantum encoding methods, and establishes cost function formulations for constrained optimization, thereby providing a unified framework for continuous quantum representations.

3.1. Complexity and Theoretical Advantages

Establishing a theoretical quantum advantage in continuous optimization first requires distinguishing between provable exact solutions and heuristic approximations [10].

In the theoretical global optimization of non-convex functions, recent work proposes a correspondence between the spectral properties of the Schrödinger operator and the mixing time of classical Langevin diffusion [22]. Real-space adiabatic algorithms (RsAA) built on this connection theoretically demonstrate polynomial-time optimization under specific, restricted structural assumptions such as block-separable functions. For strongly convex functions lacking global structure, RsAA represents a potential pathway toward polynomial complexity, though whether it can overcome the exponential bottlenecks inherent in classical methods in practical, near-term implementations remains an open question [22]. In multi-objective settings, quantum-inspired algorithms approximate the Pareto front with efficiency comparable to the classical NSGA-II algorithm across continuous search spaces [23].

In the continuous domain $\mathbb{R}^d$, minimizing a classical non-convex function $f(\mathbf{x})$ maps to the ground-state problem of a corresponding quantum system. The Hamiltonian H consists of a kinetic (Laplacian) term and a potential term defined by the objective function:

$$H = -\frac{1}{2}\nabla^2 + f(\mathbf{x}) \quad (1)$$

According to the Schrödinger equation $H\psi(\mathbf{x}) = E\psi(\mathbf{x})$, the spectral properties of this operator govern the system's evolution toward its ground state (i.e., the global minimum) [22]. Under this mapping, classical Langevin diffusion corresponds to imaginary-time evolution, while the ansatz $\psi(\mathbf{x}; \theta)$ approximates the ground-state eigenfunction.

3.2. Quantum Encoding Mechanisms

Based on this Hamiltonian formulation, encoding classical variables into quantum states fundamentally determines model expressibility. It is crucial to distinguish the optimization problem class from the physical hardware model. Continuous-variable quantum computing (CVQC) physically relies on continuous-spectrum degrees of freedom (e.g., photonic position and momentum operators) [24]. In contrast, discrete-qubit hardware requires specialized variational encoding strategies (such as angle or amplitude encoding) to embed continuous decision variables into a finite-dimensional Hilbert space.

Based on this Hamiltonian formulation, encoding classical variables $\mathbf{x}$ into quantum states fundamentally determines model expressibility [25]. Unlike quantum machine learn-

ing, which treats input data as fixed, continuous optimization requires a dynamic mapping between variational circuit parameters θ and variables $\mathbf{x}$. In continuous-variable quantum computing (CVQC), quantum states depend on operators with continuous spectra (e.g., position $\hat{x}$ and momentum $\hat{p}$). The state of a classical variable $\mathbf{x} \in \mathbb{R}^d$ can be expressed as

$$|\psi\rangle = \int_{\mathbb{R}^d} \psi(\mathbf{x})|\mathbf{x}\rangle d\mathbf{x} \quad (2)$$

While this formulation is mathematically natural, it faces practical limitations due to the discrete nature of current quantum hardware. In qubit-based architectures, the mapping is effectively realized through nonlinear transformations of the probability amplitude distribution $|\psi(\mathbf{x})|^2$ via parameterized unitary operations [26].

The choice of encoding strategy (i.e., how continuous variables are mapped to circuit parameters) directly shapes the geometry of the optimization landscape. In particular, encoding type, ansatz expressibility, and entanglement depth are strongly coupled, jointly determining the likelihood of encountering barren plateaus [20,27]. To balance expressibility and efficiency, different encoding strategies are employed. Angle encoding maps real-valued features to gate rotation angles, yielding shallow circuits and robustness to hardware noise [28]; however, its representational resolution is fundamentally limited by analog control precision.

In contrast, amplitude encoding maps normalized features directly onto the amplitudes of quantum states, enabling exponential compression of high-dimensional data using $\mathcal{O}(\log_2 N)$ qubits [29]. However, this theoretical advantage is severely bottlenecked in practice by prohibitive data-loading costs and the deep parameterized circuits required for state preparation. Furthermore, it faces high readout complexity, precision issues stemming from analog control limits on continuous amplitudes, and fundamental scalability barriers in maintaining high-fidelity superpositions across multiple qubits on near-term devices. These encoding choices ultimately define both the achievable precision and the computational complexity of subsequent gradient-based optimization [19]. As illustrated in Figure 2, unlike discrete combinatorial settings where binary variables map directly to Pauli operators, continuous optimization requires explicit embedding of variables into Hilbert space.

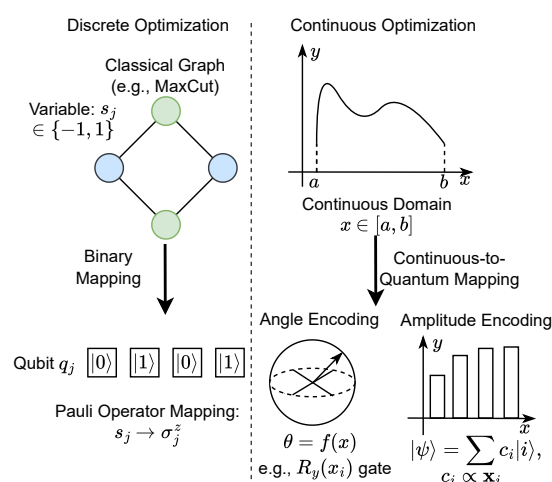


Figure 2. Comparison of quantum mapping paradigms for discrete and continuous optimization.

3.3. Cost Functions and Constraint Handling

Beyond variable encoding, continuous optimization requires transforming equality and inequality constraints into measurable quantum cost functions. Within the VQA

framework, the cost function is typically expressed as the expectation value of quantum observables [1]:

$$C(\theta) = \sum_k f_k \left(\text{Tr}[O_k U(\theta) \rho_k U^\dagger(\theta)] \right) \quad (3)$$

Conventional approaches often incorporate constraints via penalty terms [30], which can lead to severe barren plateau risks and significantly escalated measurement overhead. For problems with equality constraints $g(\mathbf{x}) = 0$ and inequality constraints $h(\mathbf{x}) \leq 0$, the classical Lagrangian is defined as $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T g(\mathbf{x}) + \boldsymbol{\mu}^T h(\mathbf{x})$. The primal–dual variational quantum framework maps this formulation to a composite operator $\hat{\mathcal{O}}_L$ consisting of multiple quantum observables [31].

In this framework, the primal state $|\psi(\theta)\rangle$ and the dual state $|\lambda(\phi)\rangle$ are prepared via separate parameterized ansatzes, transforming the optimization into a min–max problem over expectation values:

$$\min_{\theta} \max_{\phi} \langle \psi(\theta) | \otimes \langle \lambda(\phi) | \hat{\mathcal{O}}_L | \psi(\theta) \rangle \otimes | \lambda(\phi) \rangle \quad (4)$$

This operator-level construction avoids the severe landscape ill-conditioning and prohibitive sample complexity associated with large penalty terms [30]. However, practical implementation faces significant hurdles in ensuring solution feasibility. The primal–dual Lagrangian formulation necessitates dual-state preparation, leading to exponentially increased measurement overhead to evaluate composite Lagrangian operators. Navigating these min–max optimization landscapes causes severe convergence instabilities. It also creates significant execution challenges on NISQ hardware. Figure 3 illustrates this mapping from classical constrained optimization to variational quantum objectives. The resulting min–max formulation requires ansatzes capable of resolving local gradients on continuous manifolds to ensure stable convergence [32]. Consequently, designing circuit topologies that support robust continuous-variable encoding becomes central to ansatz development. Section 4 analyzes the limitations of fixed structures, while Section 5 examines dynamically adaptive architectures.

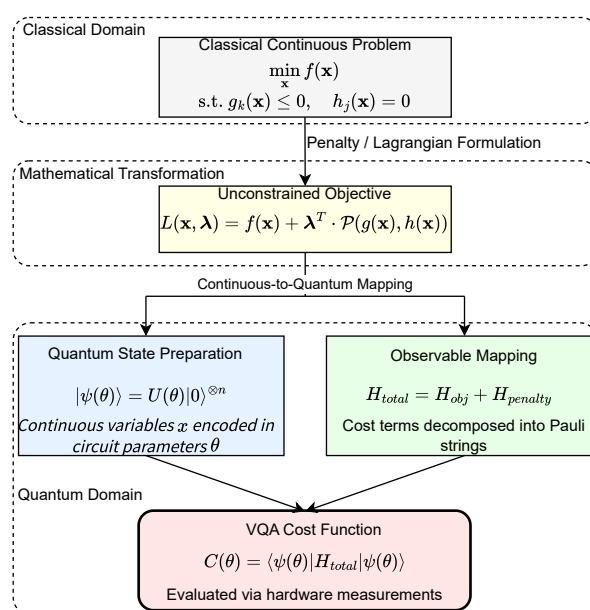


Figure 3. Schematic representation of mapping continuous constrained optimization to variational quantum algorithms (VQAs).

4. Static Ansatz Architectures: Topological Characteristics and Inherent Limitations

Variational quantum ansatz architectures for continuous optimization can be broadly categorized, based on their topology, into fixed structures and adaptive architectures, as shown in Figure 4. This section focuses on the fixed-structure paradigm. The structural characteristics of hardware-efficient ansatzes (HEAs), physics- and problem-inspired ansatzes, and custom designs are systematically analyzed, together with their intrinsic limitations in expressibility and circuit depth.

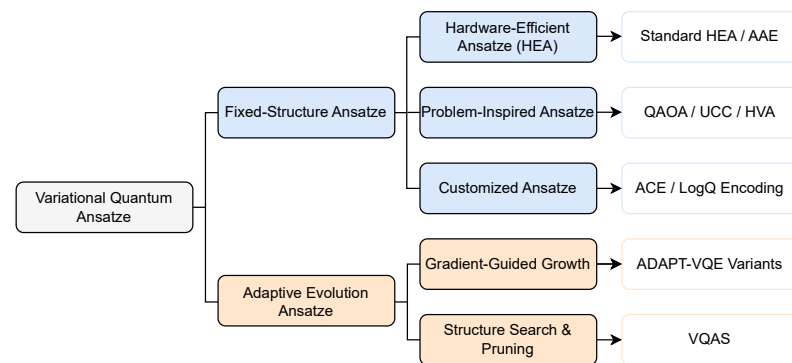


Figure 4. Taxonomy of variational ansatz architectures for continuous optimization.

To provide a structured overview of this architectural evolution, Table 2 presents a compact comparative taxonomy of all major ansatz families discussed in this review. This summary outlines their typical structures, core advantages, suitability for continuous optimization, trainability concerns (specifically distinguishing barren plateau risks), and hardware scaling limitations. Detailed quantitative analyses of their resource complexities and specific noise behaviors will be further expanded in subsequent sections.

Table 2. Compact Taxonomy and Comparative Summary of Variational Ansatz Architectures.

Ansatz Category	Circuit Structure & Hardware Scaling	Key Advantages (+) & Limitations (–)	Trainability (BP & NIBP Risk)	Suitability for Continuous Opt
Hardware-Efficient (HEA) [33]	Hardware-native gates; $\mathcal{O}(N)$ qubits	(+) Shallow, hardware-friendly; (–) Lacks physical priors	High BP risk (prone to measure concentration)	Low (relies heavily on external penalty terms)
Physics-Inspired [8,34,35]	Trotterized/Hamiltonian evolution; Polynomial to exponential depth	(+) Interpretable, symmetry-preserving; (–) Prohibitive NISQ circuit depth	Moderate BP risk; High NIBP risk at depth	High (if constraints naturally map to symmetries)
Problem-Customized [36,37]	Hardcoded topologies; $\mathcal{O}(\log N)$ qubits	(+) Explicit constraint embedding; (–) Highly task-specific	Low BP risk (restricted to shallow subspaces)	Very High (constraints inherently hardcoded)
Gradient-Driven Adaptive [38]	Iterative operator expansion; Gradient-dependent depth	(+) Compact depth; (–) Heavy commutator sampling overhead	Low BP risk; Bounded strictly by NIBPs	Moderate (good for manifold discovery, hard constraints difficult)
Geometry-Aware Adaptive [39]	Riemannian geodesic evolution; $\mathcal{O}(N^2)$ sampling overhead	(+) Avoids flat saddle points; (–) Massive metric-tensor cost	Low BP risk (saddle delay); NIBP-limited	High (for strongly non-convex continuous landscapes)
Architecture Search (VQAS) [40]	ML-driven structural search; Heavy classical compute load	(+) Prior-free, adaptable; (–) Huge classical optimization cost	Low BP risk (guided away from random regions)	High (for unstructured, hardware-aware topology search)

Note: (+) denotes advantages; (–) denotes limitations.

4.1. Hardware-Efficient Ansatzes

Early variational quantum architectures adopted static connection patterns to minimize hardware execution depth. The hardware-efficient ansatz (HEA) was introduced within the VQE framework to mitigate the substantial depth overhead introduced by Trotterization in physical simulations (e.g., quantum chemistry) [33]. Unlike problem-inspired constructions, HEAs do not rely on prior physical knowledge; instead, their structure is determined by the native gate set of the target processor [1]. The evolution operator takes the general form

$$U(\theta) = U_L(\theta_L) \cdots U_1(\theta_1) = \prod_{l=1}^L \left(\prod_m e^{-i\theta_{l,m} H_{l,m}} W_{l,m} \right) \quad (5)$$

In early hardware implementations, the circuit alternates layers of single-qubit rotations (e.g., R_x, R_y, R_z) with short-range entangling gates (e.g., CNOT or CPHASE). Each layer explores the Hilbert space via $U_{\text{layer}}(\theta) = U_{\text{ent}} \prod_{i=1}^n R_y^{(i)}(\theta_i)$, where U_{ent} employs hardware-native gates for coupling (see Figure 5). This design efficiently utilizes limited coherence time, effectively trading structured expressibility for hardware compatibility [33].

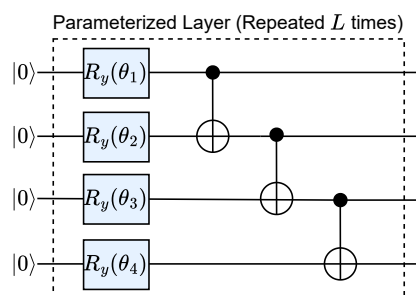


Figure 5. Schematic of a typical fixed-structure ansatz (e.g., hardware-efficient ansatz, HEA).

Extending HEAs to continuous optimization typically involves mapping real-valued features to single-qubit rotation angles. For example, in variational quantum regression for molecular energy prediction, angle encoding has been employed to embed continuous descriptors into R_y rotations. At shallow depths ($L = 2$ or 3), such HEAs can effectively capture low-complexity continuous features [28].

However, as discussed in Section 3.2, angle encoding has a strict physical resolution limit. For more complex tasks requiring high-capacity and high-fidelity representations, a transition to amplitude encoding becomes necessary. In this regime, the lack of structural bias in HEAs becomes a limiting factor: the embedding depth grows polynomially or even exponentially with feature dimension. This growth not only increases gate error accumulation but also drives the circuit toward a unitary 2-design, thereby triggering barren plateaus (BPs) [1].

To mitigate this depth growth and the associated loss of trainability, approximate amplitude encoding (AAE) replaces deterministic state preparation with a shallow variational circuit that approximates the target amplitude distribution. This variational approximation reduces circuit depth while incurring controlled fidelity loss [41]. Mitsuda et al. extended this approach to complex amplitudes (ACAE), introducing a fidelity-based objective and combining it with classical shadow techniques for efficient gradient estimation [42].

At the application level, tailoring HEA connectivity to the structure of input features can further improve performance. In medical image classification, Shoieb et al. proposed a modified HEA (MHEA) that reconfigures entangling patterns and rotation layers to better capture nonlinear spatial features [43]. HEAs are also increasingly integrated into hybrid classical-quantum pipelines. For instance, recent frameworks such as Quantum Feature

Embeddings for Graph Neural Networks (QFE-GNN) employ HEAs to embed classical node-level features into high-dimensional Hilbert spaces. Through parameterized quantum feature encoding and entangling operations, these approaches capture complex graph dependencies while linking classical structural information with quantum-enhanced representations [44]. These developments position fixed-topology HEAs as feature-mapping components within hybrid architectures. Section 4.2 summarizes their evolution and application characteristics.

4.2. Physics- and Problem-Inspired Ansatzes

For discrete optimization, QAOA introduced a physics-inspired framework based on alternating Hamiltonian evolution [8]. Extending this paradigm to continuous domains leads to CV-QAOA [34]. In this framework, the objective function $f(\mathbf{x})$ defines a potential operator $H_P = \int f(\mathbf{x})|\mathbf{x}\rangle\langle\mathbf{x}|d\mathbf{x}$, while a kinetic operator $H_M = \int p^2|p\rangle\langle p|dp$ governs exploration [34,45]. Driven by the evolution operator

$$U(\gamma, \beta) = \prod e^{-i\beta H_M} e^{-i\gamma H_P},$$

the system simulates wavepacket dynamics in a continuous energy landscape, concentrating probability mass near low-energy candidate regions [34,46]. Photonic implementations have demonstrated the feasibility of this approach on near-term hardware [46]. In practice, continuous-variable systems require truncation to a finite Fock space, which limits expressibility and introduces truncation errors [34].

Two main challenges limit CV-QAOA in practice. First, universal CV quantum computation depends on non-Gaussian gates (e.g., Kerr), which remain experimentally noisy [24]. Second, unbounded spectra can lead to instability through wavepacket diffusion [45]. Current work addresses these issues through both algorithmic and hardware-level improvements. Algorithmically, constrained evolution and complex-domain optimization techniques improve training stability. A representative example of the latter is the Complex Simultaneous Perturbation Stochastic Approximation (CSPSA) [45,47]. On the hardware side, Gaussian-only constructions and hybrid CV-DV schemes reduce the reliance on non-linear operations [48,49]. Counter-diabatic strategies further improve convergence while reducing circuit depth [50].

Beyond CV-QAOA, physics-informed ansatzes such as UCC play a central role in continuous energy optimization. Although originally formulated for electronic structure problems in quantum chemistry, the UCC architecture illustrates foundational principles of symmetry-preserving topological design that can inform continuous optimization strategies. UCC constructs unitary transformations using excitation operators,

$$U(\theta) = e^{\hat{T}(\theta) - \hat{T}^\dagger(\theta)},$$

thereby preserving physical structure [51]. However, its circuit depth scales as $\mathcal{O}(N^4)$, exceeding NISQ hardware limits. Variants such as k -UpCCGSD reduce complexity while maintaining key symmetries [52].

More generally, the Hamiltonian variational ansatz (HVA) decomposes the Hamiltonian into local terms and constructs circuits directly from them:

$$U_{\text{HVA}}(\vec{\theta}) = \prod_{l=1}^L \prod_k e^{-i\theta_{l,k} H_k}.$$

This formulation mirrors physical evolution and improves interpretability and training stability [35]. Table 3 summarizes the trade-offs of these approaches.

Table 3. Overview of Fixed-Structure Ansatzes for Continuous Optimization.

Ansatz Category	Representative Work	Algorithmic Mechanism	Key Advantages and Applications
Hardware-Efficient Ansatzes (HEA)	Standard HEA [33]	Hardware-native rotation and entangling layers	Shallow circuits; hardware-efficient compilation
	Regression HEA [28]	Angle encoding of continuous variables	Molecular energy regression
	AAE/ACAE [41,42]	Variational state preparation; classical shadows	Reduced depth amplitude encoding
	MHEA [43]	Modified entangling topologies	Enhanced local expressibility
	QFE-GNN [44]	HEA; graph neural embedding	Quantum graph feature encoding
Problem-Inspired Ansatzes	CV-QAOA [34]	Position-momentum operator alternation	Physics-mapped optimization landscape
	Wave-Packet/CSPSA [45,47]	Wave-packet constraints; complex-valued evolution	Stable continuous-variable dynamics
	Hybrid CV-DV/CD [49,50]	Hybrid control; counter-diabatic driving	Improved fidelity; accelerated convergence
	UCCSD [51]	Fermionic excitation unitaries	Electronic structure simulation
	<i>k</i> -UpCCGSD [52]	Layered excitations with symmetry constraints	Reduced circuit scaling
	Physics-driven HVA [35]	Hamiltonian decomposition	Interpretable variational dynamics
Customized Ansatzes	LogQ [36,53]	Amplitude manifold encoding	Logarithmic qubit scaling $\mathcal{O}(\log N)$
	ACE [37]	Adaptive graph-cut encoding	NISQ-friendly image segmentation

4.3. Customized Ansatzes

For problems with specific structure or large parameter spaces, general-purpose ansatzes often struggle to converge. Custom ansatzes address this challenge by incorporating prior knowledge or constraints directly into circuit design [8,36].

For the resource-efficient embedding of high-dimensional continuous variables, customized amplitude-mapping strategies have been developed. The foundational LogQ encoding scheme, originally introduced by Rančić [53], embeds an N -dimensional continuous feature vector into $\lceil \log_2 N \rceil$ qubits via $|\psi\rangle = \sum_{i=0}^{N-1} \mathbf{x}_i |i\rangle$, where $\sum_i |\mathbf{x}_i|^2 = 1$. This continuous-amplitude mapping, along with its recent extended variational applications [36], effectively reduces the spatial resource requirements from $\mathcal{O}(N)$ to $\mathcal{O}(\log N)$.

Similarly, adaptive constraint-encoding (ACE) frameworks demonstrate how structural constraints can be explicitly embedded into circuit topologies to enable efficient evaluation on NISQ devices [37]. Although frameworks like LogQ and ACE are frequently benchmarked on applied tasks such as graph feature embeddings (e.g., QFE-GNN discussed in Section 4.1) or image segmentation, their fundamental mathematical topologies explicitly manipulate continuous amplitude manifolds. Therefore, rather than being viewed merely as domain-specific applications, their specialized encoding mechanisms provide a direct structural blueprint for managing $\mathcal{O}(\log N)$ resource overhead in pure continuous optimization environments.

As these custom ansatzes scale, hardware limits on qubits and coherence times become increasingly restrictive. Circuit cutting provides a practical solution by decomposing deep circuits into smaller subcircuits that can run in parallel and be recombined afterward [54]. This approach reduces hardware demands and enables scaling to larger problems. Table 3 summarizes these architectures and their respective application regimes.

4.4. Physical Constraints and Topological Trade-Offs in Static Ansatz Architectures

All fixed-structure ansatzes face an inherent trade-off among expressibility, executable depth, and model generalization. Lacking prior knowledge, HEAs tend to over-

parameterize and approximate unitary 2-designs, thereby triggering barren plateaus (BPs) even in noise-free settings. In contrast, physics-inspired architectures (e.g., UCC and HVA, discussed in Section 4.2) suffer from prohibitive circuit depth scaling on real hardware. Custom ansatzes, as discussed in Section 4.3, alleviate some resource overhead through problem-specific structure, but their specialized topologies limit generalization in standard continuous optimization tasks. When translated into physical gate operations, these heuristic and custom architectures typically rely on Trotterization or complex graph mappings, leading to polynomial or even super-polynomial growth in the number of non-commuting operations and the overall circuit depth with system size [51].

On Noisy Intermediate-Scale Quantum (NISQ) devices, the circuit depth required for high-resolution continuous feature representation often exceeds the limits imposed by hardware coherence times and gate fidelities. Trotterized physics-inspired ansatzes (such as the $\mathcal{O}(N^4)$ scaling of the UCC ansatz [51] and the polynomial scaling of HVA [35]) frequently breach decoherence thresholds in practical deployments. This depth–noise tradeoff fundamentally limits trainability, leading to exponentially vanishing gradients, known as noise-induced barren plateaus (NIBPs) [21] (further analyzed in Section 6).

Figure 6 explicitly defines the “Static Topology Dilemma”. The conceptual framework presented in this figure is strictly grounded in rigorous theoretical bounds: unconstrained circuit growth inherently drives the optimization landscape toward expressibility-induced barren plateaus [17], while the deep physical circuits required by fixed topologies exponentially amplify gradient decay under realistic hardware errors, culminating in noise-induced barren plateaus [21]. Consequently, static designs face severe scalability bottlenecks when scaling their depth to adequately capture continuous features that encounter these unavoidable physical limits. This logical tipping point motivates the field’s transition toward gradient-guided dynamic growth paradigms. Table 4 summarizes their topological features and physical constraints, while Table 5 details their performance trade-offs and suitability for constrained continuous tasks. Furthermore, a comprehensive quantitative comparison of resource scaling, including exact qubit and depth bounds across these fixed-structure ansatzes, is provided in Table 6.

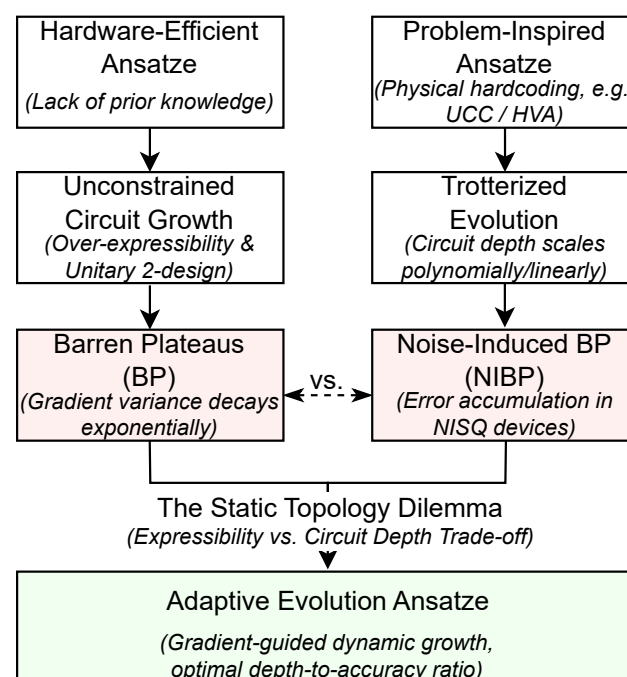


Figure 6. The inherent dilemma of fixed-structure ansatzes and the theoretical motivation for adaptive evolution.

Table 4. Comparative Summary of Topological Features and Physical Constraints of Fixed-Structure Ansatzes.

Ansatz Type	Structure	Scaling	Landscape	Noise Robustness
Hardware-Efficient (HEA [33])	Hardware-native rotations and entanglers	Polynomial/exponential	High expressibility; prone to measure concentration	Sensitive to depth-induced errors
Discrete Physics-Inspired (QAOA [8], HVA [35])	Hamiltonian-driven/Trotterized evolution	Depth-dependent	Symmetry-constrained; Trotter-depth dependent	NIBP-prone at increasing depth
Continuous Physics-Inspired (CV-QAOA [34])	Phase evolution; momentum-dependent displacement	CV-dependent	Continuous operator space; wave-packet diffusion	Sensitive to non-Gaussian gate errors
Problem-Customized (LogQ [36,53], ACE [37])	Hardcoded objectives/constraints	Logarithmic $\mathcal{O}(\log N)$	Low-dimensional manifolds; smooth landscapes	Demonstrates enhanced structural resilience primarily at shallow logarithmic depths

Table 5. Comparative Summary of Performance Trade-offs and Constraint Suitability of Fixed-Structure Ansatzes.

Ansatz Type	Advantages	Limitations	Suitability for Constrained Continuous Tasks
Hardware-Efficient (HEA [33])	Hardware-native, task-agnostic	Susceptible to BPs under deep hardware-efficient configurations lacking physical priors	Low suitability without Lagrangian penalties
Discrete Physics-Inspired (QAOA [8], HVA [35])	Interpretable; Provably convergent under noise-free state-vector assumptions	Inefficient for non-convex problems	Moderate suitability if constraints map to physical symmetries
Continuous Physics-Inspired (CV-QAOA [34])	Enables coherent CV evolution	Requires high-precision control	Moderate suitability; requires embedding constraints into continuous Hamiltonian penalties
Problem-Customized (LogQ [36,53], ACE [37])	Minimal qubit and depth requirements	Task-specific; limited generalization	High suitability; constraints are explicitly hardcoded into circuit topology

Table 6. Quantitative Resource Scaling and Hardware Deployment Constraints of Fixed-Structure Ansatzes.

Ansatz Strategy	Qubit Complexity	Circuit Depth Scaling	Gate/CNOT Overhead	Measurement/Shot Complexity	Noise Assump-tions/Constraints	Hardware Validation Status
Standard HEA (Angle Encoding) [28,33]	$\mathcal{O}(N)$ (Linear)	Fixed L layers	$\mathcal{O}(N \times L)$	$\mathcal{O}(N_{params})$ (via Parameter-shift rule)	Highly vulnerable to depth-induced decoherence	Extensively validated on physical NISQ QPUs
HEA (Amplitude Encoding)[41,42]	$\mathcal{O}(\log_2 N)$	Polynomial to Exponential	High (Requires dense multi-qubit entangling)	$\mathcal{O}(N_{params})$	Requires high-fidelity state preparation	Limited depth validated on noisy hardware
UCC (Physics-Inspired) [51]	$\mathcal{O}(N)$ (Spin orbitals)	$\mathcal{O}(N^4)$	$\mathcal{O}(N^4)$	$\mathcal{O}(N^4)$ Pauli terms to measure	NIBP-prone at increasing Trotterized depth	Limited to small molecules on real QPUs
LogQ Ansatz (Customized) [36,53]	$\lceil \log_2 N \rceil$	Shallow / Logarithmic	Low / Minimized	Reduced due to low parameter count	Inherently robust via compact encoding	Validated on physical hardware (e.g., Image segmentation)

5. Adaptive Ansatz Architectures: From Bottom-Up Growth to Architecture Search

While static ansatz architectures remain effective in specific regimes, their fixed topologies exhibit fundamental limitations when scaling to high-dimensional continuous fea-

tures—most notably excessive circuit depth and inefficient resource utilization. To overcome these constraints, ansatz design is increasingly shifting toward dynamic growth paradigms, in which circuit structure is constructed and expanded in response to the local geometry of the objective landscape during execution.

This section examines the core principles of adaptive evolution architectures, including gradient-driven operator assembly, geometry-aware adaptive strategies, and automated variational quantum architecture search (VQAS). The resource trade-offs enabled by these mechanisms are then compared.

While gradient-driven growth and architecture search form the primary axes of topological evolution, other critical techniques (such as layerwise training and noise-adaptive construction) provide indispensable, complementary mechanisms for mitigating optimization difficulty and accommodating hardware imperfections. A detailed critical evaluation of these supplementary strategies and their inherent resource trade-offs is provided in Section 5.4.

5.1. Gradient-Driven Operator Assembly

In contrast to the predefined layer stacking of static ansatzes, adaptive architectures evolve through iterative structural updates. Figure 7 illustrates the resulting differences in topology.

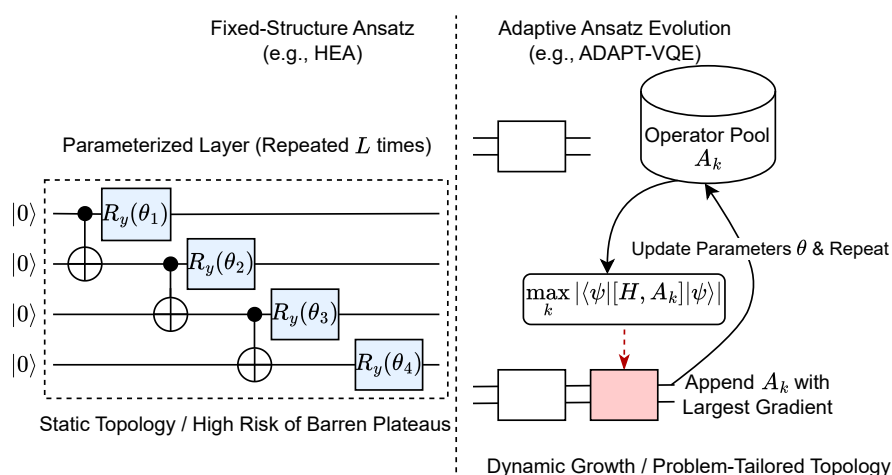


Figure 7. Structural comparison of fixed-structure and adaptive ansatzes. The red dotted arrow indicates the insertion of the operator selected from the pool (i.e., the operator with the largest gradient) into the evolving quantum circuit.

The ADAPT-VQE (Adaptive Derivative-Assembled Pseudo-Trotter VQE) framework exemplifies this paradigm by reformulating ansatz construction as a discrete, gradient-driven growth process [38]. Given a predefined operator pool $\mathcal{A} = \{A_i\}$, each iteration evaluates the gradient

$$\frac{\partial E}{\partial \theta_i} = \langle \psi | [H, A_i] | \psi \rangle$$

on the current state $|\psi\rangle$. The operator with the largest gradient magnitude is then appended to the circuit, ensuring maximal energy reduction per step.

This greedy assembly procedure incrementally builds a compact representation of the target state manifold. As shown in Figure 8, the resulting feedback loop couples circuit growth directly to local energy landscape structure [38]. Convergence is governed by the algebraic completeness of the operator pool: if the pool fails to span the relevant Lie algebra, the evolution becomes confined to a restricted subspace, leading to stalled optimization.

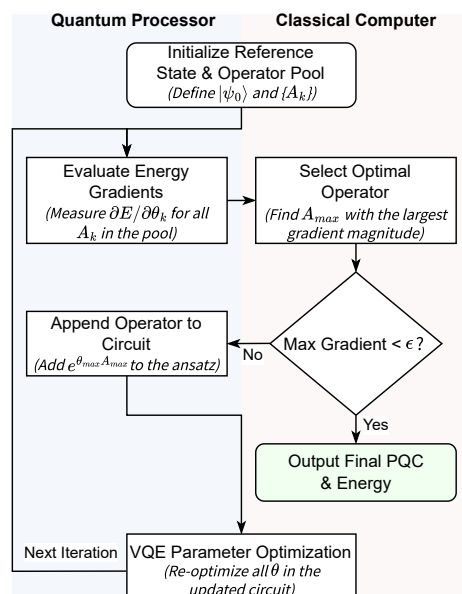


Figure 8. Schematic illustration of the iterative growth process in adaptive ansatzes (e.g., ADAPT-VQE).

By construction, this adaptive mechanism avoids premature convergence toward unitary 2-designs, thereby theoretically mitigating the onset of expressibility-induced barren plateaus through controlled expressibility growth [17,38]; however, as analyzed in subsequent sections, its practical operation remains bounded by noise-induced gradient collapse on real hardware.

Despite these advantages, hardware-software co-design principles reveal that topological decisions are directly dictated by physical realities. Limited hardware connectivity and native gate sets force the redesign and strict localization of entanglement topologies [55]. Furthermore, immense measurement costs (such as the high sampling complexity needed to evaluate commutators in adaptive VQEs [18]) and compilation overheads actively restrict theoretical expressibility and ultimately dictate the selection criteria of adaptive ansatz design. Parallelization strategies partially mitigate this bottleneck: grouping commuting observables into joint measurement schemes significantly reduces overall sampling complexity and improves gradient estimation efficiency [18]. Operator pool design further impacts efficiency. The coupled-exchange operator (CEO) pool optimizes commutation structure, yielding substantial reductions in both CNOT count and measurement overhead in molecular simulations [56].

Circuit depth remains another critical constraint. The TETRIS-ADAPT-VQE algorithm relaxes the one-operator-per-step restriction by inserting multiple commuting operators simultaneously, significantly compressing circuit depth without increasing parameter count [57]. Hierarchical growth strategies further improve robustness by structuring operator selection across sub-pools, reducing sensitivity to noise [58]. These developments reflect a broader shift from accuracy-centric to hardware-aware adaptive design.

Extensions of ADAPT-VQE incorporate physics-informed operators to improve performance on non-convex landscapes. Counter-diabatic (CD) driving introduces approximate gauge potentials into the operator pool, enhancing global optimization while reducing circuit depth [59]. Non-variational variants eliminate reliance on classical optimizers by jointly updating operator selection and parameter values based on gradient information [60]. Related approaches, such as Dynamic QAOA, further improve robustness in combinatorial settings through adaptive operator evaluation [61].

The growth process is not strictly monotonic. In practice, ADAPT-VQE can encounter flat energy landscapes, where gradients across the operator pool become uniformly small

prior to reaching the true energy minimum, thereby stalling structural updates. To address these inefficiencies, pruning-based strategies have been developed. In particular, operator reordering combined with dynamic thresholding enables the systematic removal of low-contributing operators, resulting in more compact ansatzes and improved convergence behavior without compromising energy accuracy [62].

Table 7 summarizes the main variants and optimization strategies within the ADAPT-VQE family. While gradient-driven assembly effectively controls circuit depth, local gradient information alone may fail to capture global optimization directions in highly curved manifolds. This limitation motivates the incorporation of geometric information, which is explored in the next subsection.

Table 7. Representative Improvements in ADAPT-VQE-Type Adaptive Ansatzes.

Algorithm/Mechanism	Core Improvement	Technical Approach & Expected Effect
Standard ADAPT-VQE [38]	Ansatz initialization strategy	Gradient-guided greedy operator construction
Parallel Gradient Measurement [18]	Measurement efficiency	Parallel commuting-operator measurement; reduced measurement complexity
TETRIS-ADAPT [57]	Topological compression	Parallelized disjoint operator execution; reduced circuit depth
CEO-ADAPT [56]	Operator pool refactoring	Coupled-exchange operator pool; reduced CNOT overhead (~88%)
CD-ADAPT [59]	Counter-diabatic design	Counter-diabatic operator augmentation; improved non-convex optimization
Non-variational ADAPT [60]	Optimization paradigm	Analytical gradient-based construction; optimizer-free design
Dynamic QAOA [61]	Noise robustness	Dynamic gate importance evaluation; noise-robust shallow circuits

5.2. Geometry-Aware Adaptive Mechanisms

Standard adaptive evolution algorithms rely on first-order Euclidean gradients in parameter space to guide operator assembly. This metric fails to capture the intrinsic geometry of the quantum state Hilbert space. When optimization trajectories traverse manifolds with nontrivial curvature, Euclidean gradients distort the search direction, increasing the likelihood of convergence to shallow local minima or flat saddle points [39]. Figure 9 compares optimization trajectories under Euclidean and Riemannian metrics in non-convex landscapes.

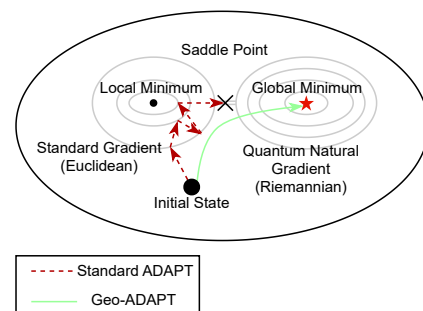


Figure 9. Conceptual comparison of optimization trajectories over a non-convex energy landscape.

To account for this geometric mismatch, the geometry-aware adaptive architecture (Geo-ADAPT-VQE) incorporates the quantum natural gradient (QNG) into the operator

selection process [39]. At each iteration, the algorithm evaluates both the first-order energy derivatives of candidate operators and the quantum Fisher information matrix (QFIM) induced by parameter perturbations. The QFIM defines the Riemannian metric tensor $\mathcal{F}$ on the parameter manifold and is equivalent to the Fubini–Study metric of the quantum state space. The natural gradient,

$$\tilde{\nabla}C(\theta) = \mathcal{F}^{-1}\nabla C(\theta),$$

therefore identifies the direction of steepest descent with respect to the underlying state-space curvature. This curvature-aware direction governs both operator selection and parameter updates.

Although Geo-ADAPT-VQE was initially validated in molecular systems with strong electron correlations (e.g., LiH, HF), the associated non-convex energy landscapes, characterized by flat saddle regions and multiple local minima, provide conceptual parallels for complex continuous optimization. Inspired by these analogous non-convex settings, geometry-aware growth rules are hypothesized to offer improved stability for continuous domains. Compared to conventional gradient-based approaches, they maintain trajectories closer to the global minimum and produce more compact ansatz circuits, reducing ground-state energy errors by approximately two orders of magnitude under identical iteration budgets [39]. These results highlight the role of Riemannian geometry as a guiding principle for structural evolution in high-dimensional non-convex optimization.

This improvement comes at the cost of increased measurement overhead. Computing the inverse QFIM requires repeated sampling of the state-space metric, where tensor construction entails pairwise inner-product evaluations between parameter derivatives. As a result, the measurement cost scales as $\mathcal{O}(N^2)$ with the number of circuit parameters, which becomes prohibitive for large-scale circuits [39]. Reducing the cost of estimating this second-order metric while preserving geometry awareness remains a central challenge for scalable implementations.

Table 8 summarizes the differences between standard gradient-driven and geometry-aware adaptive mechanisms in terms of metric structure, operator selection criteria, and landscape adaptability.

Table 8. Comparison of Underlying Features Between Gradient-Driven and Geometry-Aware Adaptive Mechanisms.

Comparison Dimension	Standard Adaptive (e.g., ADAPT-VQE) [38]	Geometry-Aware (e.g., Geo-ADAPT-VQE) [39]
Underlying Metric Space	Euclidean parameter space	Riemannian quantum state space (Fubini–Study metric)
Selection Criterion	Energy gradient-based selection	Gradient selection with QFIM geometric weighting
Evolution Trajectory	Steepest-descent parameter updates	Manifold-geodesic evolution trajectory
Landscape Adaptability	Susceptibility to local minima and flat saddles	Metric-tensor–guided search; improved saddle-point avoidance

5.3. Variational Quantum Architecture Search

While adaptive methods effectively grow circuits via local gradient steps, variational quantum architecture search (VQAS) generalizes this process by treating topology design as a global, hardware-aware meta-optimization problem. This broader search over discrete

topological spaces is often necessary for navigating complex, non-convex landscapes. In continuous optimization, objective functions typically define high-dimensional energy landscapes with pronounced non-convex structure. Designing parameterized circuits for such manifolds often requires strong physical intuition, while static architectures struggle to adequately cover the resulting feature space. Variational quantum architecture search (VQAS) casts topology design as an optimization problem, aiming to discover hardware-efficient circuits aligned with the geometry of the target manifold [40].

Formally, VQAS is typically framed as a bi-level optimization problem,

$$\min_{A \in \mathcal{A}} \left\{ \min_{\theta} C(A, \theta) \right\}.$$

During the inner-loop optimization $\min_{\theta} C(A, \theta)$, randomly initialized deep candidate architectures are prone to barren plateaus, leading to vanishing gradients that hinder effective architecture evaluation [63].

As the dimensionality of continuous variables grows, the discrete architecture space $\mathcal{A}$ expands exponentially. To address this, gradient-based quantum architecture search (GQAS) embeds discrete architectures into a continuous latent space via autoencoders, enabling efficient optimization using classical gradient descent [64]. Building on this idea, differentiable quantum architecture search (QuantumDARTS) constructs a supernet over candidate operations and employs the Gumbel–Softmax distribution to relax discrete architectural choices. In this framework, structural weights α_i define a differentiable selection probability

$$P_i = \frac{\exp(\alpha_i / \tau)}{\sum_j \exp(\alpha_j / \tau)},$$

where τ is a temperature parameter [63].

The annealing schedule of τ governs the evolution of the topology. Large τ promotes exploration over a smooth continuous space, while $\tau \rightarrow 0$ drives the distribution toward discrete one-hot selections, fixing the final architecture. This continuous-to-discrete transition enables structural features identified during optimization to transfer effectively to the resulting circuit, supporting both global topology search and local sub-circuit refinement [63].

In noisy hardware settings, identifying circuits that balance fidelity and robustness requires additional heuristics. A reinforcement learning-based framework combining Double Deep Q-Networks (DDQN) with ϵ -greedy exploration and episode termination enables efficient navigation of the architecture space toward shallow, noise-resilient solutions [65]. In parallel, evolutionary approaches such as the adaptive quantum evolutionary algorithm (AQEA) incorporate dynamic rotation updates and elitism, while introducing catastrophe mechanisms to maintain population diversity. These strategies achieve comparable predictive performance with roughly 20% fewer parameters [66].

Despite these advances, VQAS shifts the burden of complexity to the classical optimization layer. The exponential growth of the search space exposes the outer loop to the curse of dimensionality, leading to substantial training and sampling costs [63]. To mitigate this, a hierarchical search framework separates global topology exploration from local refinement, effectively reducing search complexity [67]. Further, Training-Free Quantum Architecture Search (TF-QAS) replaces expensive training with zero-cost proxy metrics, such as DAG path counts and expressibility, enabling rapid ranking of candidate circuits based solely on structural properties [68].

Although naturally framed for general variational scaling, the CutVQA framework can be extended to large-scale continuous optimization by integrating circuit cutting with architecture search via a co-design strategy. By exploiting parameter locality and en-

abling independent sub-circuit training, this approach reduces sampling and reconstruction overhead by several orders of magnitude, providing a practical pathway beyond qubit limitations [69].

While these methods alleviate barren plateaus at initialization through shallow and adaptive constructions, their scalability remains fundamentally constrained. As circuit depth increases to represent highly entangled states, accumulated hardware noise induces noise-induced barren plateaus (NIBPs), limiting further performance gains. This depth-dependent noise accumulation defines a key bottleneck for scaling dynamic ansatz architectures.

Table 9 summarizes the key characteristics of these VQAS strategies, including their driving mechanisms, search-space representations, and approaches to mitigating hardware constraints.

Table 9. Overview of Strategies and Representative Works in Variational Quantum Architecture Search (VQAS).

Search Strategy	Representative Work	Core Mechanism & Optimization	Key Challenge Addressed
Gradient-Based & Differentiable Search	QuantumDARTS [63], GQAS [64]	Gumbel-Softmax relaxation; latent-space architecture embedding	Discrete search-space exponentiality; dimensionality reduction
Reinforcement Learning & Evolutionary Search	DDQN Search [65], AQEA-QAS [66]	DDQN-guided exploration; diversity-preserving evolutionary search	Local-minima avoidance; expressibility-robustness trade-off
Hierarchical Search & Training-Free Evaluation	Hierarchical Search [67], TF-QAS [68]	Hierarchical macro-micro decomposition; training-free proxy evaluation	Training cost reduction; sampling overhead elimination
Hardware-Software Co-Design & Cut-Based Search	CutVQA [69]	Cut-aware modular search; locality-driven circuit decomposition	Scalability under qubit constraints; reduced reconstruction overhead

5.4. Layerwise Training and Noise-Adaptive Construction Strategies

Beyond gradient-driven operator assembly and automated architecture search, layerwise training and noise-adaptive construction serve as critical supplementary strategies for ansatz evolution. Layerwise training mitigates the optimization difficulty of deep parameterized circuits by starting with shallow architectures and progressively adding layers [70]. In this approach, existing parameters are either frozen or jointly fine-tuned, critically alleviating the barren plateau risks associated with random deep-circuit initialization [71]. However, this strategy introduces distinct optimization vulnerabilities. Freezing early-layer parameters inherently restricts the accessible Hilbert space, which can prematurely trap the optimization trajectory in suboptimal local minima. Furthermore, as the circuit inevitably deepens to capture high-resolution continuous features, it remains fundamentally bounded by noise-induced barren plateaus (NIBPs) on physical hardware.

Concurrently, noise-adaptive ansatz construction dynamically adjusts circuit topologies in direct response to target hardware imperfections—such as specific native gate sets, restricted coupling maps, and coherence limits—offering a pragmatic, hardware-aware alternative to pure theoretical optimization [65,69,72]. Nevertheless, the practical scalability of noise-adaptive methods is severely bottlenecked by device calibration drift. The necessity to continuously profile and update hardware noise models introduces massive classical compilation overhead. More critically, during the prolonged variational feedback loops

required for continuous optimization, the dynamically generated topologies risk becoming misaligned with the actual, fluctuating physical state of the quantum processor.

5.5. Mechanism Comparison and Resource Trade-Off Summary

Adaptive evolution mechanisms exhibit distinct trade-offs in continuous optimization. Gradient-driven approaches achieve substantial circuit depth reduction with relatively low measurement overhead. Geometry-aware methods, by incorporating Riemannian structure, offer improved convergence stability in non-convex landscapes, at the cost of an $\mathcal{O}(N^2)$ overhead for QFIM evaluation [39]. Variational quantum architecture search (VQAS) shifts topology design to a classical meta-optimization layer; while it can generate circuits tailored to hardware connectivity, this comes with significant classical computational cost [63,64].

The choice of mechanism strictly depends on the geometric properties of the energy landscape and available prior knowledge. Gradient-driven methods (e.g., ADAPT-VQE) are preferable when a meaningful physics-informed operator pool is available and local gradient estimations can efficiently guide compact circuit growth [38]. Conversely, geometry-aware methods (e.g., Geo-ADAPT) are highly suited for strongly non-convex landscapes characterized by flat saddle regions, where quantum-geometric information prevents inefficient local search directions, despite the $\mathcal{O}(N_{\text{params}}^2)$ QFIM estimation cost [39]. Finally, VQAS frameworks are most appropriate when problem-specific priors are limited and a broader, hardware-aware search over discrete circuit topologies is required, albeit at the expense of heavy classical architecture-search complexity [63]. Table 10 summarizes their technical characteristics.

Table 10. Comparative Analysis of Adaptive Evolution and Architecture Search Mechanisms.

Mechanism Type (Representative)	Evolution Feature	Parameter Scaling	Trajectory & Convergence	Noise Robustness	Advantages	Limitations
Gradient-Driven Adaptive (ADAPT [38])	Iterative operator-pool expansion	Gradient-dependent scaling	Steepest-descent growth; stagnation-prone landscape	Compact-circuit noise suppression	Compact-depth approximation; over-parameterization delay	Commutator-estimation overhead
Geometry-Aware Adaptive (Geo-ADAPT [39])	QNG-based operator screening	Curvature-dependent scaling	Riemannian-geodesic evolution; saddle-point delay	Reduced decoherence via compact circuits	Geometry-aligned evolution; improved landscape navigation	Metric-tensor computational overhead
Variational Architecture Search (VQAS [40])	ML-driven discrete/continuous search	Search-space bounded scaling	Global topology search; relaxation-based convergence	RL-guided depth-noise balancing	Prior-free design; hardware adaptability	High sampling and reconstruction overhead

Most existing adaptive frameworks, including both gradient-driven and geometry-aware approaches, are designed for unconstrained high-dimensional landscapes. However, in the presence of strict equality or inequality constraints (e.g., min–max formulations based on composite Lagrangian operators discussed in Section 3.3), current growth strategies lack mechanisms to incorporate multiplier variables or enforce hard constraints directly within operator selection [31]. As a result, dynamic ansatz construction may explore a large number of infeasible topological branches under complex engineering constraints. This limitation exposes a fundamental gap that must be addressed for practical deployment of adaptive variational architectures.

To synthesize the resource trade-offs discussed across this section, Table 11 provides a comprehensive quantitative comparison of measurement complexity, parameter scaling, and hardware validation status for these adaptive evolution architectures.

Table 11. Quantitative Resource Overhead and Measurement Complexity of Adaptive Evolution Architectures.

Adaptive Mechanism	Qubit Complexity	Circuit Depth Scaling	Parameter Growth	Measurement/Shot Complexity	Noise Assumptions/Trajectory	Hardware Validation Status
ADAPT-VQE (Standard) [38]	$\mathcal{O}(N)$	$\mathcal{O}(K)$ (Grows by 1 per iteration K)	$\mathcal{O}(K)$	Polynomial; Requires evaluating commutators across entire pool $ \mathcal{A} $ per step	Suppresses noise-free BPs; bounded by hardware coherence	Validated on physical QPUs (Small-scale)
Geo-ADAPT-VQE [39]	$\mathcal{O}(N)$	$\mathcal{O}(K)$	$\mathcal{O}(K)$	$\mathcal{O}(N_{\text{params}}^2)$ for Quantum Fisher Information Matrix (QFIM) evaluation	Avoids flat saddles via Riemannian metric; compact circuits	Noise-free/State-vector simulations only
CutVQA (Hardware-Software Co-Design) [69]	Bounded by cut sub-circuit size	Modular sub-circuits (Independent)	Fragmented	Exponential classical reconstruction overhead	Locality-driven decomposition isolates local errors	Noisy state-vector simulations/Miniapps
VQAS (e.g., QuantumDARTS) [63]	$\mathcal{O}(N)$	Dynamic/Relaxation-based	Search-space bounded	High sampling cost due to bi-level optimization & curse of dimensionality	Driven toward shallow, noise-resilient solutions	Heavily reliant on classical simulators/Limited hardware tests

6. Expressibility and Trainability: Fixed vs. Adaptive Ansatz Architectures

The intrinsic trade-off between expressibility and trainability in parameterized quantum circuits sets the fundamental limits on scaling variational algorithms to large-scale continuous optimization. This section evaluates, at the level of underlying mechanisms, how different ansatz architectures address this bottleneck.

6.1. Intrinsic Limitation of Static Topologies: Concentration of Measure and Vanishing Gradients

Barren plateaus (BPs) arising from fixed structures severely limit scalability. To quantify architectural performance, the Kullback–Leibler (KL) divergence is frequently used to compare the ensemble of states generated by a variational circuit with the uniform Haar measure, thereby characterizing expressibility [20]:

$$A^{(t)}(U) = \left\| \int dU_{\text{Haar}}(\cdot) - \int dU(\cdot) \right\| \quad (6)$$

When an ansatz exhibits high expressibility or deep entanglement, the induced state distribution in Hilbert space approaches a unitary 2-design. In high-dimensional settings, concentration of measure then drives local expectation values toward the global average, effectively flattening the energy landscape [1]. As a result, the variance of cost-function gradients decays exponentially with the number of qubits, leading to optimization stagnation. Empirically, this behavior reflects the interplay of several factors: how data encoding reshapes the feature space, how entanglement accumulates, and the expressibility of the circuit topology. Strong coupling between encoding strategies and entanglement depth accelerates convergence to a 2-design [27].

A unified analysis shows that the gradient variance scales as $\text{Var}(\nabla C) \sim \mathcal{O}(2^{-n})$ with system size n [17]. This decay can be attributed to three sources of concentration: ansatz-induced, data-encoding-induced, and problem-induced effects [17]. For continuous optimization, data encoding is often the dominant constraint for static architectures. As discussed in Sections 3.2 and 4.1, angle encoding enables shallow circuits but has a limited physical resolution. Achieving high-fidelity representations of complex con-

tinuous variables therefore often requires amplitude encoding across multiple qubits, embedding features in a high-dimensional Hilbert space. In this regime, class-conditional structure can be lost already during state preparation [17]. Increased encoding depth, combined with entangling layers, strengthens coupling effects and hastens convergence to a 2-design. Consequently, static topologies may lose trainability due to the encoding overhead before parameter optimization even begins.

At initialization, incorporating appropriate physical priors can mitigate vanishing gradients. For instance, configuring a hardware-efficient ansatz within a many-body localized (MBL) phase can prevent exponential gradient decay with increasing circuit depth [73]. Absent such priors, static topologies typically rely on carefully chosen initializations to sustain usable gradients, reflecting an inherent trade-off between expressibility and trainability.

Problem-tailored ansatzes (e.g., LogQ and ACE) encode structure directly into the circuit by restricting the accessible amplitude manifold [36,37,53]. This restriction effectively limits exploration to a reduced subspace, preventing convergence to a uniform measure and thereby locally avoiding concentration effects due to over-parameterization. However, this gain comes at the cost of generality and does not provide a universal solution for high-dimensional continuous manifolds.

6.2. Theoretical Advances of Dynamic Evolution: Decoupling Mechanisms

Adaptive architectures address the limitations of fixed topologies by breaking the tight coupling between over-parameterization and vanishing gradients.

- Mitigating encoding- and ansatz-induced measure concentration: To represent high-dimensional continuous features, static ansatzes typically rely on deep encoders and uniform entangling layers, which can drive the state ensemble toward a unitary 2-design and wash out local structure. Adaptive schemes instead grow circuits non-uniformly, guided by gradients, and confine evolution to low-dimensional submanifolds aligned with target operators [17]. This feature-driven construction limits unnecessary exploration of Hilbert space and reduces concentration effects arising from both encoding and ansatz design [1,38].
- Dynamically maintaining optimal expressibility bounds: Rather than fixing a global trade-off, adaptive evolution adjusts expressibility during optimization. At each step, it adds operators that maximize the expected gradient gain, so that increases in depth yield commensurate reductions in energy [38]. This adaptive growth improves parameter efficiency and often achieves comparable accuracy with shallower circuits than static HEAs, delaying the onset of coherence limits and NIBP effects [38,39].
- Alleviating initialization dependence: Adaptive growth typically starts from a shallow or unbiased reference state. The progressive construction of the circuit avoids the poor initializations common in deep random circuits and relaxes the requirements placed on classical optimizers [17,38].

6.3. Expressibility–Trainability Trade-Offs Under Measure Concentration

Under a unified view based on concentration of measure, different ansatz families occupy distinct positions in the expressibility–trainability landscape. Hardware-efficient ansatzes (HEAs), which maximize accessible search space, can exceed safe expressibility regimes and encounter noise-free barren plateaus [1]. Physics-inspired constructions restrict evolution to symmetry-preserving subspaces, but fixed operator sequences remain vulnerable to deep-circuit NIBPs in practice [21,51]. Problem-specific designs occupy the opposite extreme, preserving local trainability in shallow manifolds at the expense of generalization [36].

Dynamic strategies (e.g., ADAPT and VQAS) instead avoid a fixed compromise. By growing circuits on demand using gradient or geometric information, they align expressibility with the complexity of the target features. At a mathematical level, this adaptability enables partial decoupling between high-resolution representations of continuous manifolds and concentration-induced randomization [38,63]. The corresponding bounds on expressibility and trainability across architectures are summarized in Table 12.

It is crucial to rigorously separate the mechanisms driving trainability loss. Standard barren plateaus (BPs) arise algorithmically from the concentration of measure, driven by excessive expressibility, deep entanglement, global cost functions, or random initializations [1,17]. In stark contrast, noise-induced barren plateaus (NIBPs) are an artifact of physical hardware, caused directly by stochastic gate errors and accumulated decoherence [21]. These phenomena interact problematically: while gradient-guided adaptive growth or problem-specific restricted manifolds can mitigate standard BPs by avoiding over-parameterization [36,38], the resulting entanglement depth required for high-resolution continuous representations can still independently trigger NIBPs as the circuit approaches hardware coherence limits [21].

Table 12. Comparison of Theoretical Bounds in Expressibility and Trainability Across Various Variational Ansatzes.

Ansatz Type	Expressibility Space	Trainability Constraints (BP Risk)	Initialization Dependence
Hardware-Efficient Ansatzes (HEA) [33]	High expressibility; approximate Hilbert-space coverage	High barren-plateau risk (depth-induced concentration)	Strong dependence on physical initialization
Physics-Inspired Ansatzes (UCC/HVA) [51]	Restricted physical symmetry subspaces	Moderate barren-plateau risk (symmetry-constrained landscape)	Physically motivated reference-state dependence (e.g., Hartree-Fock)
Customized Ansatzes (LogQ/ACE) [37,53]	Problem-specific feature manifolds	Low barren-plateau risk (shallow-circuit structure)	Embedded problem-structure initialization
Adaptive/Architecture Search (ADAPT/VQAS) [38]	Dynamically growing low-dimensional submanifolds	Low barren-plateau risk (controlled growth; 2-design convergence delay)	Reduced initialization sensitivity via progressive construction

7. Physical Hardware Implementation and Evaluation of Variational Ansatzes

Theoretical ansatz designs ultimately require validation on real quantum hardware. This section examines the execution constraints and performance evaluation methodologies arising in deployments on physical devices. The physical constraints and noise bottlenecks associated with variational architectures on noisy intermediate-scale quantum (NISQ) systems are first analyzed. Standardized benchmarking frameworks developed to quantify the performance limits of ansatz architectures across heterogeneous hardware platforms are then reviewed. Together, these perspectives highlight the critical gap between theoretical design and practical realization.

7.1. Physical Deployment and Constraints on Noisy Quantum Devices

When evaluating the hardware readiness of these architectures, claims of NISQ applicability must be critically filtered based on the level of empirical evidence. While many adaptive architectures demonstrate promising theoretical scalability, their validation often remains confined to noise-free state-vector simulations or small-scale noisy prototypes. Genuine hardware implementation necessitates a holistic evaluation encompassing circuit depth limits, gate/CNOT overhead, real measurement costs, and the absolute requirement for quantum error mitigation (QEM) to combat decoherence [74].

During deployment on NISQ hardware, a significant gap emerges between the theoretical expressibility of ansatzes and their physical realizability. Hardware-efficient ansatzes (HEAs) were originally designed to map directly onto native gate sets, thereby minimizing circuit depth [33]. However, when extended to complex continuous-variable tasks, limited qubit connectivity and finite coherence times impose strict constraints on achievable circuit depth and entanglement structure [55]. As a result, the theoretical advantages of expressive ansatzes often cannot be fully realized in practice.

In addition to structural constraints, hardware noise fundamentally alters the optimization landscape. Stochastic gate errors and decoherence distort the cost function and introduce bias and variance into gradient estimates used by classical optimizers [75]. This noise-induced perturbation not only slows convergence but can also redirect optimization trajectories toward suboptimal regions. Moreover, calibration drift and device-specific imperfections lead to temporal instability, reducing reproducibility across repeated runs [76]. These effects highlight that variational optimization on hardware is inherently a stochastic and non-stationary process.

A central manifestation of these limitations is the emergence of noise-induced barren plateaus (NIBPs). Under realistic noise models, even linear increases in circuit depth can induce exponential decay in gradient magnitudes due to accumulated decoherence [21]. This phenomenon establishes a fundamental trade-off between circuit expressibility and trainability under realistic noise conditions. Importantly, this degradation cannot be fully mitigated by ansatz design alone. Consequently, quantum error mitigation (QEM) techniques are increasingly integrated into variational workflows to improve the accuracy of expectation value estimation and enhance optimization stability [74]. These observations suggest that effective ansatz deployment requires joint consideration of circuit design, noise characteristics, and mitigation strategies.

7.2. Standardized Benchmarking of Ansatz Architectures

To systematically evaluate ansatz performance on real hardware, several standardized benchmarking frameworks have been developed. A robust benchmarking protocol must holistically evaluate multi-level metrics. Low-level structural suites (e.g., QASMBench [77]) are essential to measure circuit depth and parameter counts. Mid-level execution metrics, such as convergence reliability, gradient variance, and noise sensitivity, can be effectively evaluated using uncertainty-aware frameworks [76]. High-level system metrics are equally critical: quantum miniapp frameworks [78] evaluate solution quality and scalability under realistic workloads, while asynchronous dataflow tools (e.g., Tierkreis [79]) help benchmark the number of required measurements and overall hardware compatibility.

At the application level, benchmark pipelines incorporating domain-specific datasets further assess the effectiveness of variational ansatzes in end-to-end tasks [80,81]. These evaluations account for system-level factors such as scheduling, resource allocation, and hybrid classical–quantum interaction overhead. Collectively, these benchmarking efforts establish a principled foundation for comparing adaptive and fixed ansatz architectures under realistic hardware constraints.

Overall, existing evaluation methodologies are evolving from static structural metrics toward dynamic, noise-aware, and application-driven benchmarks. This shift reflects a broader transition in variational quantum computing: from idealized algorithmic analysis to hardware-grounded performance characterization. Establishing unified and reproducible benchmarking standards remains essential for guiding the design of next-generation ansatz architectures and for fairly assessing progress toward practical quantum advantage.

Furthermore, benchmarking criteria must shift away from discrete-focused metrics and be dynamically tailored to capture continuous manifold features [80,81]. To ensure

reproducibility and fair evaluation, future studies should adopt the following standardized protocols:

- **Benchmark Problem Specification:** Selection of continuous non-convex test functions (e.g., Rastrigin, Ackley) mapped to appropriate hardware scales [80,81].
- **Encoding & Ansatz Details:** Explicit reporting of the continuous-to-quantum mapping strategy (e.g., angle vs. amplitude encoding) and complete circuit topology.
- **Circuit Resources:** Quantification of exact qubit requirements, parameterized gate counts, and CNOT overhead.
- **Optimization & Measurement Settings:** Transparent disclosure of classical optimizer hyperparameters, learning rate schedules, and explicit measurement shot-complexity counts [6].
- **Performance Metrics & Noise Models:** Reporting of empirical KL divergence [20], gradient variance [17], and mandatory evaluation under combined hardware-specific noise channels (e.g., depolarizing and amplitude damping) [21].
- **Reproducibility Protocols:** Mandatory inclusion of classical optimization baselines, documentation of random seeds, performance averaged over multiple independent runs, and open-source availability of code and raw data.

8. Conclusions and Outlook

Despite rapid architectural advancements, several fundamental barriers hinder practical deployment. First, the theoretical scalability limits of expressive ansatzes conflict directly with hardware coherence times. Second, all discussed architectures exhibit extreme sensitivity to hardware noise, wherein NIBPs dominate gradient scaling regardless of topological design. Third, adaptive methods (e.g., ADAPT-VQE variants) impose prohibitive measurement shot costs for continuous commutator evaluation. Fourth, limited physical hardware connectivity restricts the implementation of complex entangling topologies without excessive SWAP-gate overhead. Finally, the absence of standardized continuous benchmarks across heterogeneous hardware platforms remains a critical gap in rigorously validating computational advantages.

This review systematically examined the evolution of variational quantum ansatz architectures for continuous optimization. In continuous parameter spaces, efficient mappings between classical variables and quantum states, together with compatible circuit topologies, are essential for achieving computational advantage. Static ansatzes with fixed structures, however, face an inherent tension between expressibility and trainability under hardware constraints. Increasing circuit depth to improve representation power often induces barren plateaus (BPs), arising from redundant parameterization and excessive expressibility. These limitations have motivated a shift toward dynamically adaptive architectures.

Synthesizing this progression, four core architectural principles emerge that directly address the specific bottlenecks of continuous quantum optimization. First, regarding quantum representation mechanisms, circuit topology must be strictly co-designed with continuous data encoding. This coupling is essential to control the effective dimensionality of the Hilbert space, prevent early measure concentration, and mathematically accommodate equality and inequality constraints. Second, the analysis of static architectures reveals an inescapable topological dilemma: pre-defined structures, such as hardware-efficient and physics-inspired ansatzes, cannot simultaneously scale expressibility and maintain trainability, inevitably triggering barren plateaus in high-dimensional non-convex landscapes. Third, dynamically adaptive architectures overcome these structural rigidities by shifting to on-demand topological evolution. By exploiting gradient-driven assembly, Riemannian geometry-aware metrics (e.g., the quantum natural gradient), and automated architecture search (VQAS), these mechanisms systematically align circuit growth directly

with the underlying manifold curvature. Fourth, theoretical and physical limits reveal that architectural adaptability must ultimately be evaluated against measure concentration and hardware reality. The intrinsic trade-off between expressibility and trainability dictates that noise-induced barren plateaus (NIBPs) and gate fidelities constitute the fundamental limits for executing any deep ansatz on NISQ devices.

To address the limitations of static designs, next-generation approaches increasingly combine hardware–software co-design with physics-informed strategies. At the algorithmic level, gradient-based construction methods are evolving toward geometry-aware and higher-order manifold-based frameworks. By incorporating metrics such as the quantum natural gradient, these methods align parameter updates with the curvature of the quantum state manifold, thereby improving optimization stability. At the architecture-search level, integrating classical machine learning into variational quantum architecture search (VQAS) enables data-driven circuit generation and meta-learning of optimal topologies. From a theoretical standpoint, an important objective is to embed strict optimization constraints directly into operator-level formulations while maintaining manageable qubit and measurement overheads.

Despite these advances, several fundamental challenges persist. Noise-induced barren plateaus (NIBPs) indicate that architectural adaptation alone cannot prevent gradient collapse caused by accumulated hardware errors. This hardware bottleneck highlights a critical limitation of NISQ-era variational frameworks. In addition, scalable encoding of high-dimensional continuous variables remains costly, particularly for amplitude-based representations. While schemes like LogQ offer a theoretical blueprint for logarithmic spatial compression, their practical realization on NISQ hardware is fundamentally constrained by extreme requirements for analog control precision, which serves as a major bottleneck for achieving genuine quantum advantage in pure continuous optimization. Constraint-aware ansatz construction is still underdeveloped, limiting applicability to real-world engineering problems. Furthermore, the lack of standardized benchmarking protocols for continuous optimization under realistic noise models hinders fair comparison across methods.

Addressing these challenges requires coordinated progress across multiple layers of the stack. In the near term, practical deployment will rely on improved encoding strategies, noise-aware compilation, and quantum error mitigation (QEM) techniques to extend effective circuit depth. Recent empirical evidence suggests that ansatz-level error mitigation, such as zero-noise extrapolation (ZNE) applied to parameterized unitaries, provides a complementary layer to structural optimization, as the performance of even the most noise-aware ansatz is fundamentally bounded by the readout fidelity of the underlying hardware. Concurrently, the development of benchmark suites incorporating hardware noise, constraint complexity, and large-scale continuous tasks will be essential for rigorous evaluation. In the longer term, integrating variational frameworks with fault-tolerant quantum computing (FTQC) and quantum error correction (QEC) will be necessary to overcome the fundamental limitations of NISQ devices.

By combining adaptive circuit construction, geometry-aware optimization, and hardware–algorithm co-design, next-generation variational quantum algorithms explore potential pathways toward advantage, highlighting the strict conditions under which quantum speedup may become plausible. However, realizing scalable and noise-resilient quantum optimization remains strictly contingent upon overcoming unresolved challenges, specifically the high overhead of continuous data encoding, the pervasive threat of NIBPs, prohibitive measurement costs, and the establishment of universally accepted benchmarking standards.

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